

Answer any five (05) sets of the following questions

1. (a) Differentiate between ethics, law, and professional responsibility with suitable examples from the computing field. 5
(b) Discuss any two historical cases of ethical failures in technology and explain the lessons learned from them. 5
(c) Explain the principles of software engineering ethics, highlighting the issues related to professional responsibility. Discuss how the ACM/IEEE Code of Ethics guides ethical decision-making in software engineering practices. 4
2. (a) Compare and contrast Utilitarianism and Deontology (Kantian ethics) as ethical frameworks. How would each approach handle a software privacy dilemma? 4
(b) Explain Virtue Ethics and describe how it can guide ethical decision-making for software engineers. 6
(c) Define ethical relativism and ethical absolutism. How can cultural differences influence ethical judgments in global software development projects? 4
3. (a) Explain the concept of professional negligence in computing. How does it differ from an honest technical mistake? 4
(b) What is whistleblowing? Discuss the ethical considerations and challenges a computing professional might face when deciding to blow the whistle. 6
(c) The ACM Code of Ethics emphasizes both "Contribute to society and human well-being" and "Respect privacy." Describe a realistic scenario where these two principles might come into conflict for a data scientist. How might the code guide a professional in resolving this tension? 4
4. (a) What is digital manipulation? Discuss ethical issues related to deepfakes and synthetic media. 5
(b) How should ethical conflicts between intellectual property rights and open-source innovation be resolved in the ICT sector? 5
(c) Explain the ethical and social consequences of misinformation and fake news propagated through digital platforms. 4
5. (a) Discuss the concept of informed consent in personal data collection. What ethical principles should organisations follow when collecting user data? 7
(b) What are the ethical concerns associated with surveillance technologies? Explain with examples how these technologies can be misused. 4
(c) Briefly explain the ethical importance of documentation and transparency in software projects. 3
6. (a) What is algorithmic bias? Explain its causes and suggest measures to ensure fairness in AI systems. 5
(b) Discuss the importance of transparency and explainability in AI. Who should be held accountable when an autonomous system causes harm? 4
(c) Discuss the role of ethical leadership in IT organizations. How can leaders foster an ethical culture within their teams? 5
7. (a) Differentiate between ethical hacking and cybercrime. Under what conditions is hacking considered ethical? 6
(b) Explain the concept of responsible disclosure in cybersecurity. What ethical guidelines should a security researcher follow when discovering a vulnerability? 2
(c) Discuss the ethical implications of employee monitoring and workplace surveillance using digital technologies. 6
8. (a) What is the digital divide? Discuss its social implications and suggest measures to bridge this gap. 6
(b) Explain the concept of green computing. How can computing professionals contribute to environmental sustainability? 2
(c) What is a conflict of interest in the IT workplace? Provide examples and explain how professionals should ethically handle such situations. 6